

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Please provide a brief explanation of your answer.

$$x_1 \approx 0.00, x_2 \approx 1.00$$

The problem is to minimise total treatment time $T = x_1 + x_2$ subject to $x_1 + a_2x_2 = 1$ with $x_1, x_2 \geq 0$.

If $a_2 > 1$, the optimal solution is $x_1 = 0, x_2 = 1/a_2$.

If $a_2 < 1$, it is $x_1 = 1, x_2 = 0$.

If $a_2 = 1$, any (x_1, x_2) on the line segment is optimal.

The code defines a_2 as the limit of `Compute_a_2(n)` as $n \rightarrow \infty$.

Evaluating the limit:

- $f(n)$ sums $(1/2)^k$ from $k = 3$ to $n + 2$ (which converges to $1/4$) and minuscule terms involving $72^{-10^{24}}$.
- The limit of `Compute_a_2(n)` is $1 + 2 \cdot 72^{-10^{24}} - \text{negligible} > 1$.
Thus a_2 is strictly greater than 1.

Therefore the unique optimal solution is $x_1^* = 0, x_2^* = 1/a_2$.

Since $a_2 = 1 + \varepsilon$ with $\varepsilon \approx 2 \cdot 72^{-10^{24}}$ (unimaginably small), $1/a_2 \approx 1 - \varepsilon$, which equals $1.00\dots$ to far more than two decimal places.

The values $x_1 = 0.00, x_2 = 1.00$ approximate the minimiser within an ℓ^∞ error well below 0.005.